Week 5: Bridging the Linguistic Gap – NLP Integration and Final System Synthesis

1. Introduction and Objectives for Week 5

Week 4 was a major technical milestone: the system was able to use a Hybrid CNN-LSTM architecture and Connectionist Temporal Classification (CTC) Loss to process continuous, unsegmented video streams to generate an ordered sequence of ASL glosses. Nonetheless, although the model might correctly predict single signs, the raw output was not yet a fluent translation to a user who would not sign.

Week 5 had the main goal of integrating Stage 4: Natural Language Processing (NLP) Integration. It was centered on the linguistic alignment between ASL syntax and English grammar, on the integration of the separate modules into a unified work of real-time performance, and on the final performance analysis with the original standards of the project.

2. The Linguistic Challenge: ASL Glosses vs. English Syntax

According to what is written in the first project proposal, American Sign Language is a unique language that has its own system of grammar. It is not word-to-word manual reflection of the English. A word-for-word translation of ASL tends to produce a glossing, or series of root words, which, although semantically comprehensible, do not have the prepositions, indicators of tense, and word order to form a fluent English sentence.

As an illustration, a typical English sentence like, I am going to the store can be signed conceptually as, Store I go. The system needed a translation layer with the ability to make contextual reasoning in order to translate these raw sequences into natural language such that the project would meet its goal of generating fluent grammatical sound text.

3. Implementing the LLM Translation Layer

To resolve this syntax disparity, I integrated a Large Language Model (LLM) API as the final stage of the pipeline. This module acts as a bridge between the decoded CTC output and the final user interface.

The implementation process involves:

- **Sequence Buffering:** As the CTC decoder collapses the video frames into a sequence of signs, these words are held in a short-term buffer.
- **Contextual Refinement:** The raw ASL glosses are passed through the LLM API to contextualize the signs.
- **Generative Translation:** The LLM restructures the sequence into a fluent, grammatically valid English sentence, effectively inferring the necessary English components that are not always represented by individual hand signs.

4. Full Pipeline Integration and Real-Time Optimization

With all four stages complete, the latter half of the week was dedicated to full system synthesis. The final application integrates the following functional loop:

1. **MediaPipe Holistic:** Captures the webcam stream and extracts 1,662 3D landmarks per frame.
2. **Encoder-Decoder LSTM:** Processes the spatial data to track the dynamic movement of signs.
3. **CTC Decoding:** Collapses the input frames into a sequence of raw words without requiring manual segmentation.
4. **NLP Refinement:** Converts the sequence into structured English sentences for display on the real-time user interface.

To ensure the system met the stringent latency requirements necessary for live translation, I utilized a sliding-window implementation. This approach manages video buffering to ensure the entire pipeline maintains a minimum of 30 frames per second.

5. Final Evaluation and Results

To conclude the development cycle, I evaluated the integrated system against the performance metrics established in my literature review. The final outcomes aligned with the expected deliverables:

- **Model Performance:** The Hybrid CNN-LSTM model demonstrated a strong capability for predicting continuous ASL glosses.
- **User Interface:** The functional, real-time interface successfully captures webcam input and displays translated text with minimal latency.
- **System Robustness:** Initial testing confirmed the model's environmental robustness, maintaining accuracy across varied testing conditions.

6. Conclusion

In the last five weeks, this project has been developed as a theoretical framework to a working, real-time Continuous ASL Translation tool. The system offers a more realistic and open bridge to communication by surpassing the individual word recognition and addressing the challenges of co-articulation and ASL syntax. This expedition has proven that the fusion of spatial feature extraction, time modeling and generative NLP is an effective and feasible way forward in assistive technology.